AgentCore as the Sandbox in Agentic RL

Running slime's tool calls, checkers and agent loops on Amazon Bedrock AgentCore

Kaige Yang
Senior Applied Scientist, AWS

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02 AgentCore foundations

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03 Using AgentCore in slime

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06 Other demos

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07 Conclusion

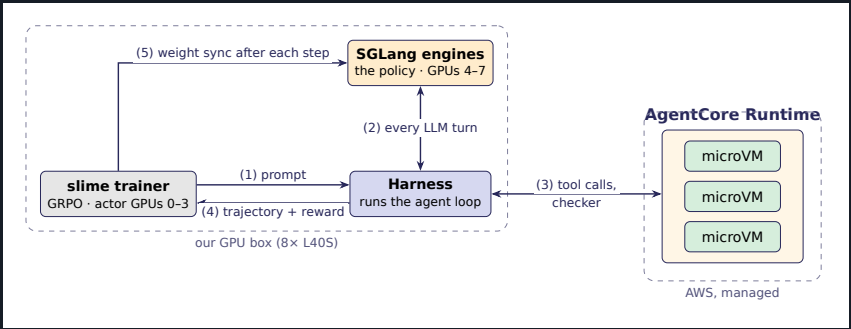
DeepSeek DSec vs AgentCore, and takeaways

Appendix (from slide 30): references.

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The sandbox is the reward source: four tiers close one on-policy loop

Figure 1 **Agentic RL: trainer, inference, harness, sandbox**

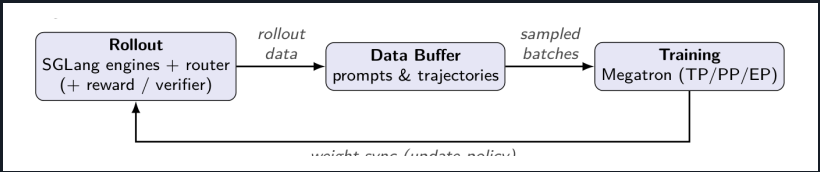


- **Trainer** (slime: Megatron, GRPO) updates the policy; **inference** (SGLang engines) serves it.
- **Harness** runs the agent loop; a proxy records the exact `token_ids` + logprobs (**TITO**), so training stays on-policy.
- **Sandbox** (AgentCore) runs the tool calls and the checker: **the sandbox is the reward source**.

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Each episode becomes one loss-masked Sample, and rollout runs async with training

Figure 2 slime pipeline: rollout → data buffer → training → weight sync



1. **Seed:** the harness opens a sandbox session and writes the task's files.
2. **Act → observe:** policy turn → tool call runs in the sandbox → output fed back, for several turns.
3. **Check:** a hidden checker runs in the sandbox → binary reward (RLVR, no LLM judge).
4. **Pack:** one loss-masked `Sample` per episode → data buffer.
5. **Train + sync:** a GRPO step, weights pushed to SGLang; rollout runs **async** with training.

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The same loop code runs on either substrate; only the Runtime ARN differs

- **Runtime:** your image, serverless, **per-session isolated compute**, called with `InvokeAgentRuntime` (SigV4).
- **Code Interpreter:** a managed CPU sandbox, no image to build.
- Same loop code on both substrates; only the Runtime ARN differs. **Both demos use microVMs.**

Table 1 **microVM vs Instance substrates**

	microVM (serverless)	**Instance** (EC2 via a capacity provider)
Compute	own guest kernel, **CPU only**	m5 (CPU) · g6 / g5 (**GPU**, whole cards, no MIG)
Image	**arm64**, $\leq$ ~**2 GB**	**amd64**, instance disk
Session length	$\leq$ 8 h	$\leq$ 14 days

Source: AgentCore substrate limits as stated in the project deck (`docs/slides/agentcore_slime_deck.md`) and report (`docs/report/agentcore_slime_report.md`); checked 2026-09-25.

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A cold launch costs 0.77 s to minutes; a reused or pre-opened session opens in milliseconds

- **1 : N : 1.** Runtime = one image, one ARN; session = one `runtimeSessionId` = one microVM / instance. **Runtimes track images, not tasks.**
- **Launch:** no “create session” API; the first invoke under a new id cold-starts the compute.
- **End:** idle timeout, max lifetime, or `StopRuntimeSession`.

Table 2 **Measured launch cost by substrate**

Substrate	Cold launch (measured)	Reused open
Code Interpreter	p50 **1.83 s**, **every episode**	–
Own Runtime, microVM	≈ **0.77 s** (stdlib) · ≈ **15 s** (447 MB)	pre-open wait **5 ms**
Instance `m5` · `g6`	≈ **60–80 s** · ≈ **123–207 s**	0.12–0.33 s

Source: Report `docs/report/agentcore_slime_report.md`: `DATASET=libs` smoke (`start_session` p50, 2026-09-25); warm-start probe `warmstart_20260925_100103` (2026-09-25); GPU sweep (2026-09-21).

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One RL step of 16 prompts $\times$ 8 rollouts needs 16 Runtimes and 128 fresh microVMs

Batch: 16 prompts, **each needs its own image**; GRPO samples **8 rollouts per prompt**.

Table 3 One batch mapped onto Runtimes, sessions and microVMs

RL concept	AgentCore object	Count for this batch
Prompt env	1 image $\rightarrow$ 1 **Runtime** (ARN)	**16 Runtimes**, deployed once, reused every step
Rollout	1 **session** (session id)	16 $\times$ 8 = **128 sessions**, new every step
Its compute	1 **microVM** per session	**128 microVMs** in flight, stopped at episode end
GRPO group	8 sessions on **one** Runtime	8 separate microVMs, nothing shared

- **AgentCore never sees prompts or rewards**, only an ARN + session id; the harness maps them.
- **Limits:** 128 opens at 25 /s $\geq$ **5 s**/step; 128 $\ll$ 5000 concurrent; Runtimes = **distinct images in the dataset** ($\leq$ 1000).
- **Cost:** 128 cold boots per step (0.8–15 s each) unless pre-opened (slide 16).

Source: Worked example; quotas from AgentCore Service Quotas (us-west-2, 2026-09-26), boot times from the warm-start probe (2026-09-25).

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Silent idle reclaim, the 25/s open rate and the 1000-Runtime cap shape the design

Table 4 **AgentCore limits and how we handle them**

Limit	Value	What we do
Idle timeout	15 min default (we set 3600 s)	reclaim is **silent** (empty sandbox) $\rightarrow$ heartbeat
Max lifetime	microVM **8 h** · Instance 14 d	re-seed long episodes
New sessions	**25 /s** (adj.)	caps the cold-start burst $\rightarrow$ pre-open ahead
Concurrent sessions	**5000** (adj.)	one per in-flight episode
Code Interpreter	**30 /s** start and invoke	$\approx$ 75 episodes in flight
Runtimes per account	**1000**; create $\leq$ 50 /s	$\leq$ ~1000 custom images (Terminal-Bench ~229)
Image	microVM **2 GB**, arm64	bigger $\rightarrow$ slim it or use an Instance
Per request	**15 min** · 100 MB	long `exec` runs async, polled
Privileges	**no Linux capabilities**	no per-episode uid / cgroup inside one session

Source: AgentCore Service Quotas, this account, us-west-2 (2026-09-26); project deck `agentcore_slime_deck.md` (2026-09-25).

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slime needs only its two existing hooks to use AgentCore, so there is no fork

`--custom-generate-function-path` → `generate.generate` (dispatches the episode, records turns); `--custom-rm-path` → the reward.

```
async def generate(args, sample, sampling_params): # slime hook
    task, pool = task_from(sample), harness_pool.get_pool(args)
    async def episode(worker): # a free dispatch worker pulls it
        async with pool.sandbox(key, open_session) as (sb, wait, slot): # sandbox first
            with pool.engine() as (e, engine): # then the least-loaded engine
                return await mini_claude_episode.run(..., sb, engine) # loop runs IN sb
    return await pool.run(episode) # ONE loss-masked Sample
```

- Each sandbox op = one `invoke_agent_runtime(ARN, session_id)` in a thread → **N episodes concurrently**.
- `__aexit__` ends the session when the episode returns: one teardown path.
- `SANDBOX=agentcore-runtime` → your own Runtime: the agent loop runs **in the session** and dials the trainer back, which the Code Interpreter can't.

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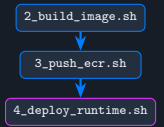
Deploying a Runtime takes three steps, and only the dial-back path needs ingress

Build, IAM, wire

1. **Build + deploy:** image (**arm64** microVM / **amd64** Instance) → ECR → `CreateAgentRuntime`. The image serves `sandbox_server_agent.py`: `write_file` / `exec` / `read_file`.
2. **IAM:** an execution role (ECR pull, logs) + an operator role with `InvokeAgentRuntime`. Instances also need `CreateCapacityProvider`.
3. **Wire** (`deploy/runtime.env`): `AGENTCORE_SANDBOX_RUNTIME_ARN` · `..._ARNS` (round-robin) · `..._ARN_BY_IMAGE` (per image).

Networking: pull, don't expose

SigV4 invoke, **no inbound port**. Only dial-back (in-session harness → trainer) needs ingress: one SG→SG rule under `networkMode=VPC`, plus interface endpoints (`ecr.api`, `ecr.dkr`, `s3`, `logs`, `sts`, `bedrock-agentcore`); a missing one fails create with a **null reason**.

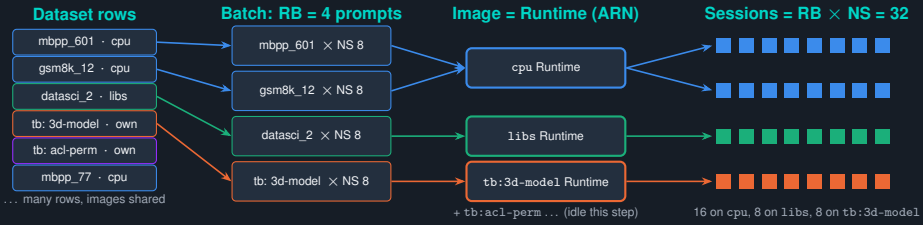


Each demo wraps the steps as these scripts.

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Every row names a task and an image; each image is one Runtime; RB × NS sets sessions per step

Figure 3 CPU tasks + Terminal-Bench in one batch (`DATASET=mbpp,terminalbench`): row → Runtime → sessions



- **Row = task + image.** Rows may share one: every MBPP / GSM8K row → `cpu`; pandas / scipy / sklearn → `libs`; each Terminal-Bench task → its own image. The NS samples of a prompt share it.
- **Image = one Runtime**, via the map image → ARN. An unmapped image is an **error, never a fallback**. `cpu` + `libs` = 2 Runtimes; Terminal-Bench adds one per task (166 usable of 218 deployed).
- **Sessions per step = RB × NS** (default 4 × 8 = **32**), one fresh session each, spread over the Runtimes the batch touches.

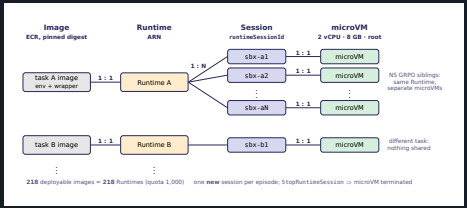
Source: `code/demos/demo_cpu_tasks` (row schema, `image_routing.py`) and the Terminal-Bench pool map (`tasks/arn_by_image.json`), 2026-09-28.

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Of 229 Terminal-Bench tasks, 166 are usable, each on its own image and Runtime

- **~229 terminal / SWE tasks, each its own environment**: its own OS, packages and a deliberately broken state. Reward = 1 only if every hidden `pytest` passes.
- The environment *is* the task → **one arm64 image + one Runtime per task**, routed by task id.
- Tasks mutate the whole OS (apt, `/etc`, services) → **a fresh microVM per episode**; the session is the isolation boundary.

Figure 4 **Image 1:1 Runtime 1:N session 1:1 microVM**



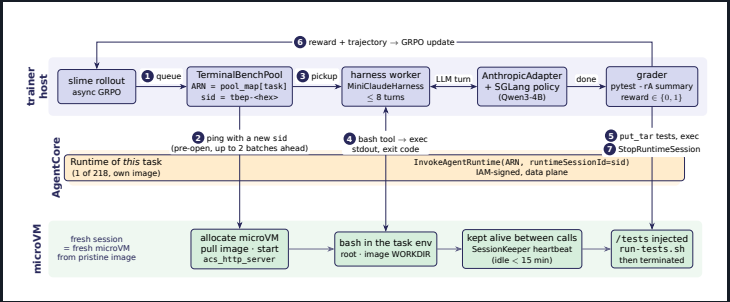
229 tasks → **218 deployable** (arm64, ≤ 2 GB) → **166 oracle-usable** → **135 train / 31 eval.**

Source: Terminal-Bench task audit in `code/demos/demo_cpu_tasks/terminalbench` (`tasks/arn_by_image.json`); project deck, 2026-09-25.

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Every episode boots a fresh microVM: the first invoke with a new session id is the boot

Figure 5 One Terminal-Bench episode: route → fresh session → agent loop → grade → stop



- **Route → fresh session → agent loop → grade → stop.** The first invoke with a new `runtimeSessionId` **is** the boot.
- Hidden tests are injected only **after** the agent finishes; `StopRuntimeSession` terminates the microVM.
- One step = RB 4 × NS 8 = **32 episodes = 32 fresh microVMs**.

Source: `demo_cpu_tasks DATASET=terminalbench` configuration (RB 4, NS 8); project deck, 2026-09-25.

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Serial cold boots alone would cost about 8 minutes per step, and slime cannot hide them

Table 5 Terminal-Bench key challenges

Challenge	Why it hurts
Cold start every episode	≈ 15 s for a 447 MB image; 32 per step ≈ **8 min** of serial boot if on the critical path
Bursty opens	all 8 GRPO siblings of a task start at once; account cap **25 new sessions/s**
slime can't hide it	only `CONC` × `engines` = 8 episodes visible; step N+1 drawn only after step N; slime knows nothing about sandboxes
Silent idle reclaim	a session idle ≈ 15 min is reclaimed (same id, empty sandbox), and weight sync can be that long
Broken / oversized tasks	7 don't build on arm64, 4 exceed the 2 GB cap, 47 fail their own reference solution
Runtime cap	1000 Runtimes per account → at most ~1000 images

Source: Terminal-Bench task audit and first AWS smoke (2026-09-23); AgentCore Service Quotas (2026-09-26).

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Each challenge has a fix: fresh VMs, strict routing, oracle checks, pre-open, heartbeat, retries

Table 6 Terminal-Bench challenges and our solutions

Challenge	Solution
Isolation vs reuse	**Fresh microVM per episode**; reuse + reset rejected (OS state leaks into the reward)
Many images	**One image + Runtime per task**, strict routing (`tasks/arn_by_image.json`; unmapped = error)
Broken tasks	**Oracle check**: untouched env must FAIL, reference must PASS $\rightarrow$ **166 usable**
Cold start	**Look-ahead pre-open**: boot the next 2 batches' sessions before slime asks (slide 16)
Idle reclaim	**Heartbeat** (`SessionKeeper`) pings every open session (slide 17)
Infra failures	**Retry on a new id** ($3\times$); logged as `infra_error`, never scored as a wrong answer

Source: `code/demos/demo_cpu_tasks/terminalbench`; project deck, 2026-09-25.

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Pre-opening cut the p50 open wait to $\leq$ 5 ms and the makespan by 15–20 %

1. **Look-ahead data source** (`tb_lookahead_source.py`, a `-data-source-path` plugin) replays slime's deterministic prompt order → the next 2 rollouts' prompts. **No slime change.**
2. **Bounded window:** `TB_PREOPEN_BATCHES=2` → **P = 64** sessions booted ahead, pooled per task; each claim starts the next boot. 1 → 32, 0 → off.
3. **Sizing:** boot hidden when P · E / W $\geq$ boot time; beyond that is idle cost. Tune by `open_wait_ms` ($\gg$ 0 → raise) vs `preopen_idle_ms` (the price).

Table 7 **Warm-start probe (24 episodes, batches of 6): cold open vs pre-open**

Backend	Mode	Wait p50	Waited	Makespan
own Runtime	cold → **preopen**	774 → **5 ms**	24 → 8 /24	31.9 → 25.5 s (−**20 %**)
Code Interpreter	cold → **preopen**	1715 → **0 ms**	24 → 6 /24	28.7 → 24.3 s (−**15 %**)

Source: Warm-start probe `demo_cpu_tasks/terminalbench/warmstart/logs/warmstart_20260925_100103.*` (2026-09-25, us-west-2).

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With heartbeat on, own-Runtime sessions survived a long idle 3/3; without it, 0/3

- During weight sync no op reaches the sandbox → your own Runtime **silently reclaims** idle sessions (same id, empty sandbox, no error).
- `SessionKeeper` pings any session idle past `fraction × timeout` with `exec("true")`.
- **On by default** for own Runtimes (`HEARTBEAT=0` = off), idle timeout 3600 s as backstop. Code Interpreter: set a long session timeout instead.

Table 8 **Heartbeat vs silent idle reclaim**

Backend	Heartbeat	Survived 150 s idle (60 s timeout)	Resume
own Runtime	on	**3/3**	104 ms, files intact
own Runtime	off	0/3	**silently reclaimed**
Code Interpreter	on / off	0/3	expires at `sessionTimeoutSeconds`

Source: Warm-start probe `warmstart_20260925_100103` (2026-09-25, us-west-2), heartbeat leg.

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Pre-open fully hid the boot (p95 wait 0), but solves are sparse at 4 of 382

Table 9 **Terminal-Bench smoke results (2026-09-23)**

Metric	Value
Episodes / infra errors	382, each on its own fresh microVM / **0**
Boot on the critical path (`open_wait_ms`)	**p50 0** · **p95 0**: pre-open hides it fully
Session open	p50 0.66 s · **100 % warm hits** · 0 retries
Pre-opened VM idle p50	**7.3 min** for a 21 s episode → ≈ 95 % idle → try `TB_PREOPEN_BATCHES=1`
Solves	**4** / **382** train, eval 0.0; 42 % pass ≥ 1 hidden test → sparse; next: curriculum / partial credit

In plain English: the sandbox side works: no infra errors and no visible boot wait. The open problem is the reward signal, not AgentCore.

Source: First Terminal-Bench AWS smoke, `demo_cpu_tasks DATASET=terminalbench` (2026-09-23).

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The harness must scale with inference and the sandbox, or it silently becomes the ceiling

- **Use case covered:** scale an agentic-RL rollout by sizing **inference, harness and sandbox independently**; the decoupled harness tier (`harness_pool.py`) is the **default in every demo**, and `SCALE=1` turns all three up together (4 : 256 : 256, DeepScaleR on a stdlib image; `demo_cpu_scale_lanes` was retired into this preset, 2026-09-28).
- **Why this demo:** (1) the harness must scale **along with** SGLang and the sandbox, or a few coroutines silently become the ceiling; (2) it should scale **with the sandbox, not as part of the trainer**: the agent loop runs in the episode's session, as in DeepSeek DSec.
- **Design considerations:** **throughput = min(inference, harness, sandbox)**; each tier is its own pool, joined per episode (slide 20). Pinned together (worker = engine = microVM), the slowest pin sets the pace.

Source: `code/demos/demo_cpu_tasks` (`SCALE=1` preset), checked 2026-09-28.

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Each tier binds on a different limit, and each has its own signal to read

Table 10 The three tiers to scale

Tier	Scales with	Limited by	Signal that it binds
Inference (SGLang)	GPUs / engines	decode throughput	`engine_inflight`
Harness (agent loops)	Python concurrency	event loop, client pools	`qdepth`
Sandbox (AgentCore)	sessions in flight	**25 new/s**, 5000 concurrent, cost	`sandbox_wait_ms`

In plain English: the rollout goes only as fast as its slowest tier. Find which one is full before buying more of anything.

Source: `demo_cpu_tasks` harness tier (`harness_pool.py`); AgentCore Service Quotas (2026-09-26).

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SCALE=1 runs 4 engines : 256 dispatchers : 256 sessions, opening at 20/s under the 25/s quota

Defaults in every demo: **W = RB × NS** dispatchers (32), a fresh session per episode, least-loaded engine. `SCALE=1` values below; client pools sized to 256 (boto3 default 10, **a silent cap**). Shrink one tier to make it the ceiling, then read its signal.

Table 11 **Tier sizing in** `demo_cpu_tasks SCALE=1`

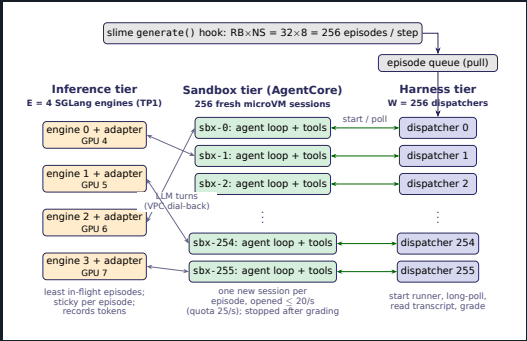
Tier	Default	Knob	Why this number
SGLang	**4 engines**, TP1, GPUs 4–7	`GPUS_PER_ENGINE` (1; default 2)	Qwen3-4B fits one L40S; other 4 GPUs train
Harness	**256** dispatchers	`HARNESS_WORKERS` (256)	= one step's batch, RB 32 × NS 8 = 256; CONC = 64 per engine
Sandbox	**256** fresh sessions in flight	`SANDBOX_SESSIONS` (0 = fresh per episode; N = N pooled)	one per in-flight episode
Open rate	**20 /s**	`SANDBOX_OPEN_RATE`	under the 25 /s quota → 256 open in ≈ 13 s

Source: `code/demos/demo_cpu_tasks SCALE=1` preset (2026-09-28); AgentCore Service Quotas (2026-09-26).

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Nothing is pinned: each episode gets a fresh session and the least-loaded engine

Figure 6 Harness tier at `SCALE=1`: 4 engines : 256 dispatchers : 256 fresh sessions, agent loop in the session



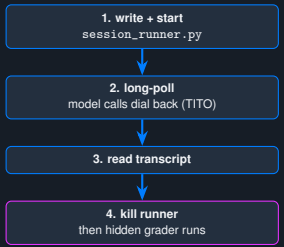
Per episode: a free dispatcher pulls it, opens a **fresh session** (sandbox first), starts the agent loop **inside** it; the loop dials back to the **least-loaded engine** (sticky for its turns). Nothing is pinned: this replaces the old hash pinning and per-engine workers.

Source: `demo_cpu_tasks` harness tier, every demo; shown at `SCALE=1` (2026-09-28).

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The agent loop runs inside the sandbox session, so the harness tier scales with the sandbox

- **One agent loop per sandbox, in the same session**, in every demo, like DSec (only exception: `demo_cpu_branching` runs the real Claude Code CLI in its microVM).
- **Per episode:** write + start `session_runner.py` → long-poll → read transcript → **kill the runner, then the hidden grader** runs (the loop can't touch its reward).
- **Model calls dial back** (VPC, SG→SG) to the engine's `AnthropicAdapter`, which records the exact tokens (TITO).
- **So the harness tier scales with the sandbox tier**: trainer-side dispatchers only start and poll.
- **No fallback:** no `python3` or no path back → `HarnessPlacementError`; the loop never moves into the trainer.



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One command runs SCALE=1; isolation defaults to a fresh microVM per episode

```
cd code/demos/demo_cpu_tasks
python3 harness_scaling_demo.py # offline min() sweep, seconds
SCALE=1 bash 5_run.sh # after steps 1-4: DATASET=deepscaler RB=32 NS=8
                                    # E=4 (TP1) HARNESS_WORKERS=256 MODE=A CONC=64 -> S=256
```

- **Isolation:** default = **fresh microVM per episode**, stopped after grading.
- `SANDBOX_SESSIONS=N` (opt-in) pools N microVMs per image with a fresh `ep-<hex>` workdir only (`/tmp`, packages not reset → fine for DeepScaleR, not for OS-mutating tasks).

Source: `code/demos/demo_cpu_tasks` (`5_run.sh`, `harness_scaling_demo.py`); checked 2026-09-28.

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The 256-worker smoke ran 592 episodes and raw reward rose from 0.31 to 0.37

Table 12 Harness-scaling results

Run	Result
Session mode, live	verified per episode: good solution **1.0**, cheat **0.0**
Live pool check (64 : 64, 192 episodes)	**1280 ops ok**; engines 48 / 48 / 47 / 49
256-worker RL smoke (`SCALE=1`)	**592 episodes**, 2 steps × 256; `raw_reward` **0.31** → **0.37**, eval/deepscaler **0.27**
Earlier 4 × 4 × 4 lanes	eval **0.358**

In plain English: at 256 concurrent episodes the load spread evenly over the 4 engines (48 / 48 / 47 / 49), and raw reward rose over the 2 steps.

Source: 256-worker RL smoke, logs in `code/demos/demo_cpu_tasks/logs/scale/` (2026-09-28); live pool check and lanes run from the project deck (2026-09-25).

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Five demo folders share the same hook, harness tier and step scripts, across microVMs, Instances, GPUs and EKS

3 main folders + an EKS twin + one experimental, each standalone in `code/demos/` with the same `1_prepare_data.sh` … `6_cleanup.sh` steps and the same harness tier.

Table 13 **Demo folders in the codebase**

Folder	What it shows
`demo_cpu_tasks`	literature tasks (MBPP, GSM8K, MATH-500, DeepScaleR, HumanEval, SQL, HotpotQA, xLAM, MCP, mixes) on a generic stdlib Runtime, plus **Terminal-Bench** (`DATASET=terminalbench`, one image + Runtime per task, slide 12; was `demo_cpu_tasks DATASET=terminalbench`); `SCALE=1` is the harness-scaling run (slide 19)
`demo_cpu_swebench`	one image + Runtime per **SWE-bench Verified** instance on **m5 Instances** (images $>$ 2 GB microVM cap); reuses the Terminal-Bench pool
`demo_cpu_swebench_eks`	the same SWE-bench workload on **EKS**, one fresh **gVisor** pod per episode; cluster not created yet
`demo_gpu_farm`	**KernelBench** CUDA RL; **one GPU instance serving many sessions** via a Ray farm (`FARM=fractional` 0.5 + 0.5 or `whole`), local or on a g6 (`LOCAL_GPU=0`); trusted-only
`demo_cpu_branching` *(experimental)*	the **real Claude Code CLI** as harness; one episode fans out into many samples

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DSec outscales AgentCore's 25 opens/s; pre-open and heartbeat fill the gaps

Figure 7 DeepSeek DSec vs AgentCore as an RL sandbox

	DSec (DeepSeek)	AgentCore (AWS)
Scale		
Creation rate	$>$ 5,000 sandboxes /s	**25** /s new Runtime sessions
Running at once	380K	**5,000** Runtime · **1,000** Code Interpreter sessions
Environments	11K bases $\times$ 100K workspaces, layered at start time	one image per Runtime; max **1,000** Runtimes per account
Images and start time		
Image delivery	lazy loading: only the 4–13 % of the image a sandbox touches is fetched	**full image pulled per cold boot**: $\approx$ 0.8 s stdlib, $\approx$ 15 s at 447 MB, grows with size; **2 GB cap** on microVMs. **Ours:** no image on Code Interpreter (pandas/scipy/sklearn/torch-cpu preinstalled); install heavy deps at boot to fit 2 GB; $>$ 2 GB on a Runtime Instance (60–80 s m5, $\approx$ 207 s g6 boot, paid once via warm-shared session)
Hiding the cold boot	not needed: lazy loading keeps pull time low; pre-warming rejected (the bottleneck is I/O volume, not latency)	**pre-open (ours)**: while the current batch runs, the next 1–2 batches' sandboxes are opened (image pulled, microVM booted); ready-wait p50 774 ms $\rightarrow$ **5 ms**, makespan -20 %
Lifecycle		
Sandbox lifetime	p50 15–17 min, p99 $>$ 3 h	**15-min idle timeout** by default; max 8 h (microVM), 14 days (Runtime Instance)
Trainer stall	pause/resume, state kept	**no pause/resume, no state kept**; **heartbeat (ours)** keeps the sandbox alive
Architecture and safety		
Agent loop	outside the GPU training job	**outside the trainer too (ours)**: harness + tools in the same Runtime session, one loop per sandbox (`HARNESS_WHERE=session`, default on own Runtimes)
Reward hacking	AppArmor + per-task network allowlists	microVM boundary; Landlock jail and no TCP on shared pools; VPC mode for the network

green adjustable (quota or per-Runtime setting) **orange** our mitigation **red** AgentCore limit / missing

- **Scale:** $>$ 5,000 sandboxes/s and 380K at once (DSec) vs **25 /s** and 5,000 Runtime sessions (AgentCore).
- **Images:** lazy loading vs a full pull per cold boot; ours hides it with **pre-open**.
- **Stalls:** pause/resume vs none; ours bridges it with **heartbeat**.
- **Agent loop:** outside the trainer in both.

Green = adjustable, orange = our mitigation, red = AgentCore limit / missing. Full-size figure: slide 32.

Source: DSec: arXiv 2609.22978 (V3.2 $\rightarrow$ V4.1, 3M sandboxes/day). AgentCore: Service Quotas, this account, us-west-2, 2026-09-26.

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CONCLUSION

DSec's sandbox lifetimes match our idle timeout, so heartbeat and lifetime metrics are now default

- **Their median sandbox lives about as long as our idle timeout** → heartbeat on by default; idle timeout 3600 s.
- **New metrics, every demo:** `perf/sandbox_ready_span_s`, `perf/sandbox_open_peak_per_s` (vs 25/s), per-session lifetime + longest idle gap (`<log>.sessions.csv`).
- **Terminal-Bench, re-parsed:** 414 sessions, lifetime p50 **471 s** / p95 **1841 s**; 55 outlived 900 s (kept by heartbeat); peak **36 opens/s**, **0 throttles**.
- **Not exposed by AgentCore:** image layers, session pause, thousands of creates/s.
- **Copied:** agent loop in the sandbox session (slide 23). **Still to copy:** per-task network allowlist.

Source: Terminal-Bench smoke (2026-09-23) re-parsed with the new session metrics; DSec arXiv 2609.22978; report Ch. 7.

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Start generic, add images only when needed, and scale each tier on its own signal

- **Start on one generic stdlib Runtime**: most literature tasks need no custom image (the Code Interpreter has the packages, but can't host the in-session agent loop).
- **Add an image only when the environment differs**; route by `task.image`, AgentCore sees only an image and a session.
- **Hide boots with pre-open, bridge stalls with heartbeat** on your own Runtimes (slides 16–17).
- **Scale the tiers independently**: the harness tier with the sandbox, not in the trainer; diagnose the binding tier (`qdepth` / `sandbox_wait_ms` / `engine_inflight`) before adding capacity.
- **Isolation = the session**: fresh microVM when tasks mutate the OS; pooled microVM + fresh workdir for workdir-only tasks.

Sources for every number: slide 31.

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Appendix

References · DSec comparison, full size

References: every number traces to a project run, log or document

Table 14 Sources used in this deck

Source	Used for
`docs/slides/agentcore_slime_deck.md` (2026-09-25)	source deck for all content
`docs/report/agentcore_slime_report.md`	launch costs, probe details, DSec comparison, GPU sweep (2026-09-21)
`demo_cpu_tasks/terminalbench/warmstart/logs/`	pre-open and heartbeat probe `warmstart_20260925_100103.*` (2026-09-25, us-west-2)
`code/demos/demo_cpu_tasks/terminalbench`	task audit (229 → 166), first AWS smoke (2026-09-23), `tb_lookahead_source.py`, `tasks/arn_by_image.json`
`code/demos/demo_cpu_tasks/logs/scale/`	256-worker SCALE=1 RL smoke (2026-09-28)
`code/demos/demo_cpu_tasks`	`harness_pool.py`, `session_runner.py`, `harness_scaling_demo.py`
`docs/agentcore-networking.md`	pull-vs-dial-back networking, VPC endpoints
AgentCore Service Quotas, us-west-2 (2026-09-26)	session rate, concurrency, Runtime and image limits
DSec, arXiv 2609.22978	DeepSeek sandbox figures (V3.2 → V4.1, 3M sandboxes/day)

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Backup: AgentCore trails DSec on scale and images; loop placement matches

	DSec (DeepSeek)	AgentCore (AWS)
Scale		
Creation rate	> 5,000 sandboxes /s	**25 /s** new Runtime sessions
Running at once	380K	**5,000** Runtime · **1,000** Code Interpreter sessions
Environments	11K bases × 100K workspaces, layered at start time	one image per Runtime; max **1,000** Runtimes per account
Images and start time		
Image delivery	lazy loading: only the 4–13 % of the image a sandbox touches is fetched	**full image pulled per cold boot**: ≈ 0.8 s stdlib, ≈ 15 s at 447 MB, grows with size; **2 GB cap** on microVMs. **Ours:** no image on Code Interpreter (pandas/scipy/sklearn/torch-cpu preinstalled); install heavy deps at boot to fit 2 GB; > 2 GB on a Runtime Instance (60–80 s m5, ≈ 207 s g5 boot, paid once via warm-shared session)
Hiding the cold boot	not needed: lazy loading keeps pull time low; pre-warming rejected (the bottleneck is I/O volume, not latency)	**pre-open (ours)**: while the current batch runs, the next 1–2 batches' sandboxes are opened (image pulled, microVM booted); ready-wait p50 774 ms → **5 ms**, makespan −20 %
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green adjustable (quota or per-Runtime setting) **orange** our mitigation **red** AgentCore limit / missing

Source: DSec: arXiv 2609.22978 (V3.2 ⟶ V4.1, 3M sandboxes/day). AgentCore: Service Quotas, this account, us-west-2, 2026-09-26.

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Thank you

Questions?

Kaige Yang · Senior Applied Scientist, AWS